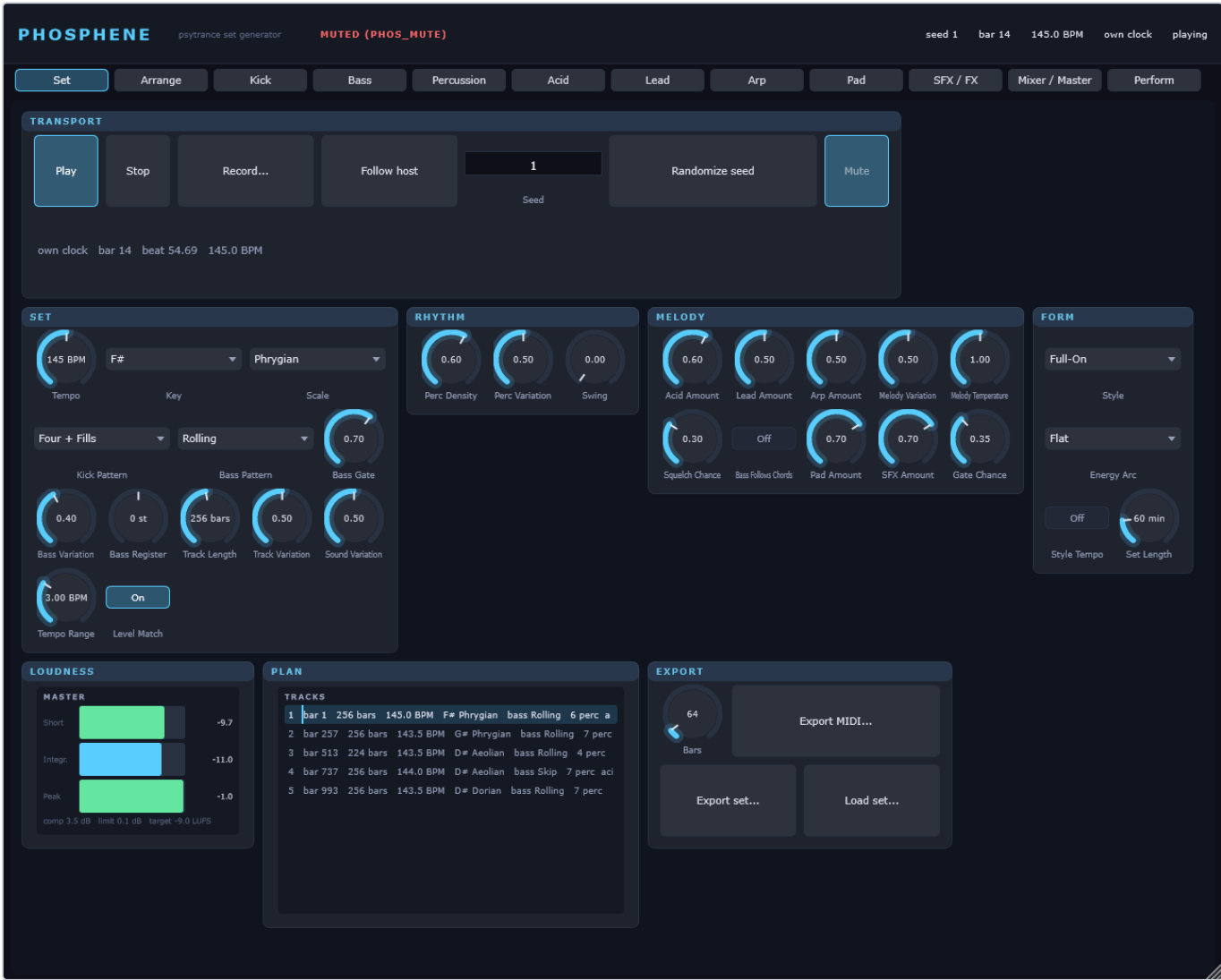


Phosphene

Manual · version 0.1.0



614 parameters · 12 tabs · 4 perform macros · every picture in this manual is the plugin drawing itself at design size

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1. About

Phosphene composes and plays complete psytrance sets. You give it a seed, a style and an energy arc; it writes the tracks -- their keys, tempi, forms, patterns and sounds -- and synthesises them in real time. There is no sample library and no loop pool: every sound is made by the engine, and every note is decided by the composer.

Two ideas run through the whole instrument. The first is that the same seed always gives the same set: bar 3000 is bar 3000 whether the night was played from the start or joined in the middle, which is what makes a set something you can save in seven lines of text and hand to someone else. The second is that the composer never writes a knob. It writes offsets from the knobs, so with every offset at zero the instrument sounds exactly as the panel says, and a variation keeps its musical size whatever you have set the ranges to.

The panel has a tab per generator, plus four that are about the set rather than the sound: Set, Arrange, Mixer / Master and Perform. Every control on every page is generated from the engine's own parameter tables, and so is this manual.

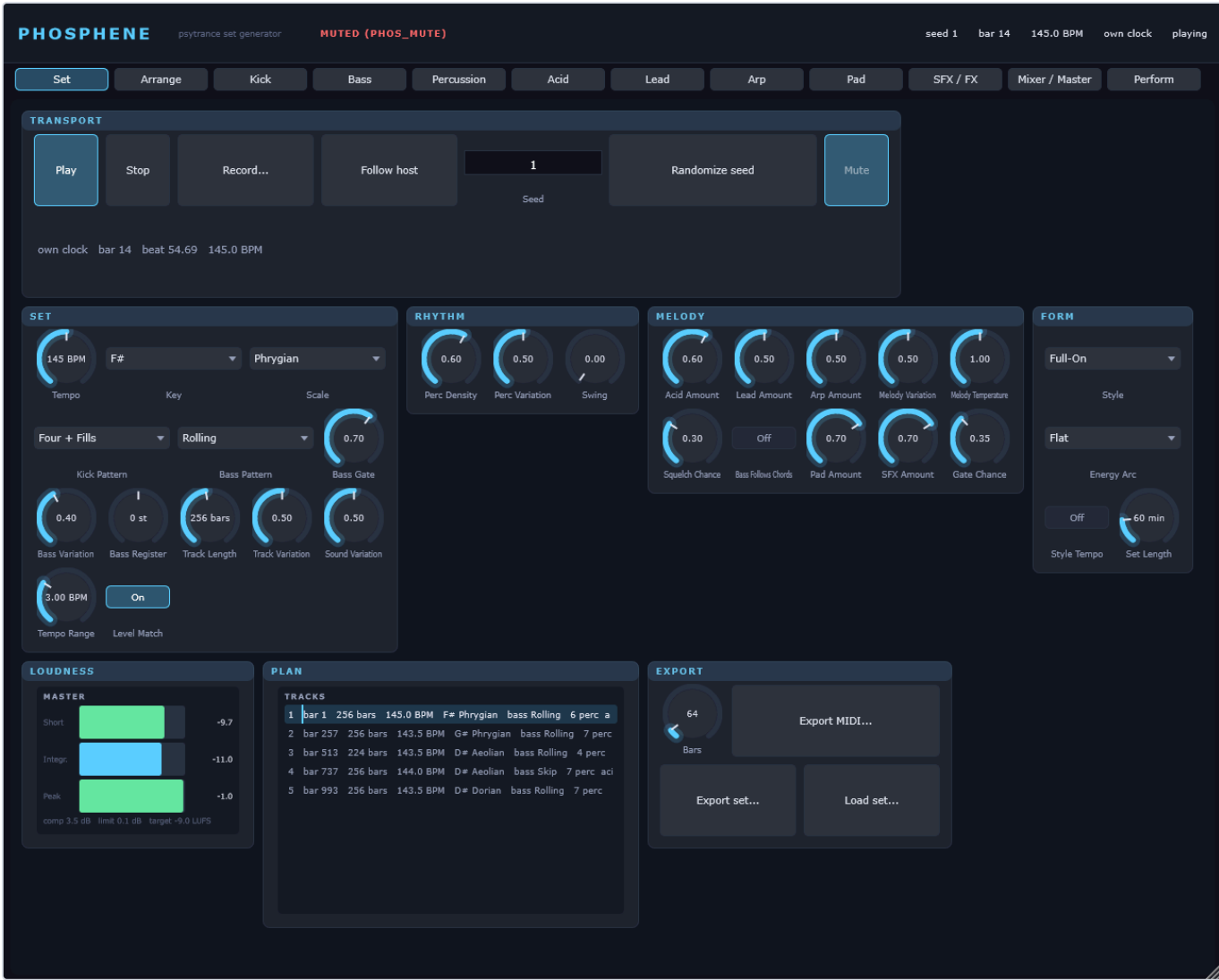
2. Set

The Set tab is where a night begins. The seed names the set; Randomize seed draws another one. Play and Stop run Phosphene's own clock, and Follow host hands the transport to the DAW instead -- under a host the tempo knob follows the host's tempo and the play head follows its position.

Starting a set is not instant. Planning a track measures its loudness by rendering two bars of it, so the first sound arrives about two seconds after the button; the header says "planning" rather than going quiet without explanation. In the standalone this has usually happened before you press anything, because the composer thread starts working when the window opens.

The composer's own knobs are here: how long a track is, how far the key and tempo may wander, how much of each voice there is, which style profile weights the grammar, which dramaturgy the energy arc follows and how long the whole set should be. The first track of a set plays exactly these knobs; variation starts with the second.

Export MIDI writes the score of the first N bars as a Standard MIDI File with a named track per part, section markers and the tempo map. Export set writes a .phosset -- the seed, the style, the arc, the knobs that differ from their defaults, and every lock and reroll -- which phos_render will play back byte for byte.



The Set tab.

SET

Control	Key	Range	Default
Tempo	compose.bpm	100 .. 190 BPM	145
Key	compose.key	C C# D D# E F F# G G# A A# B	F#
Scale	compose.scale	Aeolian Phrygian Harmonic Minor Phrygian Dominant Double Harmonic Dorian	Phrygian
Kick Pattern	compose.kick_pattern	Four Four + Fills Off	Four + Fills
Bass Pattern	compose.bass_pattern	Rolling Gallop Skip Offbeat Triplet	Rolling
Bass Gate	compose.bass_gate	0.2 .. 1	0.7
Bass Variation	compose.bass_variation	0 .. 1	0.4
Bass Register	compose.bass_register	-12 .. 12 st	0
Track Length	compose.trackBars	32 .. 512 bars	256
Track Variation	compose.track_variation	0 .. 1	0.5
Sound Variation	compose.sound_variation	0 .. 1	0.5
Tempo Range	compose.tempo_range	0 .. 10 BPM	3
Level Match	compose.level_match	off / on	on

RHYTHM

Control	Key	Range	Default
Perc Density	compose.perc_density	0 .. 1	0.6
Perc Variation	compose.perc_variation	0 .. 1	0.5
Swing	compose.swing	0 .. 0.3	0

MELODY

Control	Key	Range	Default
Acid Amount	compose.acid_amount	0 .. 1	0.6
Lead Amount	compose.lead_amount	0 .. 1	0.5
Arp Amount	compose.arp_amount	0 .. 1	0.5
Melody Variation	compose.melody_variation	0 .. 1	0.5
Melody Temperature	compose.melody_temperature	0.5 .. 2	1
Squelch Chance	compose.squelch_chance	0 .. 1	0.3
Bass Follows Chords	compose.bass_follows_chords	off / on	off
Pad Amount	compose.pad_amount	0 .. 1	0.7
SFX Amount	compose.sfx_amount	0 .. 1	0.7
Gate Chance	compose.gate_chance	0 .. 1	0.35

FORM

Control	Key	Range	Default
Style	compose.style	Goa Full-On Progressive Dark Forest Hi-Tech	Full-On
Energy Arc	compose.arc	Warm-up Peak-Time Morning Closing Flat	Flat
Style Tempo	compose.style_tempo	off / on	off
Set Length	compose.set_minutes	10 .. 300 min	60

3. Arrange

The Arrange tab is the whole planned set on one page, and the curation loop that goes with it.

The strip across the top is the night: every planned track a block in the order it plays, the energy arc drawn over it, the play head in it. Underneath, each track gets a row of its own in which its own form fills the full width, so a section stays readable no matter how long the set is. The colours are the section categories: intro and outro grey, groove green, buildup amber, the pre-drop break violet, the drop red, the breakdown blue. Clicking any block jumps there.

The two small glyphs on every track and every section are the answer to the oldest complaint about generators, which is that they are boring. The padlock freezes that unit on the seed it has; the die draws it again. Because a locked unit keeps its old seed and nothing later reads anything an earlier one decided, rerolling one track leaves every other track bit-identical -- so you can keep what you like and reroll what you do not, one section at a time, until the set is yours. Reroll the set draws the whole journey again and leaves the locked tracks exactly where they were.

PHOSPHENE psytrance set generator **MUTED (PHOS_MUTE)** seed 1 bar 14 145.0 BPM own clock playing

Set **Arrange** Kick Bass Percussion Acid Lead Arp Pad SFX / FX Mixer / Master Perform

SET TIMELINE

ARRANGE 5 tracks 1248 bars planned set 60 min

1 F# Phrygian 145.0 BPM Intro Groove Drop Break Drop Outro

2 G# Phrygian 143.5 BPM Groove Drop Break Drop Break Drop

3 D# Aeolian 143.5 BPM Drop Break Drop Drop Drop Outro

4 D# Aeolian 144.0 BPM Intro Groove Drop Break Drop Break Drop Outro

5 D# Dorian 143.5 BPM Intro Groove Drop Break Drop Drop Outro

the strip is the night to scale, every row is one track filling the width - click a block to jump - padlock freezes a track or a section on its seed - die draws it again (a locked one does not move)

CURATION

Reroll the set Reroll this track Lock this track Clear all locks

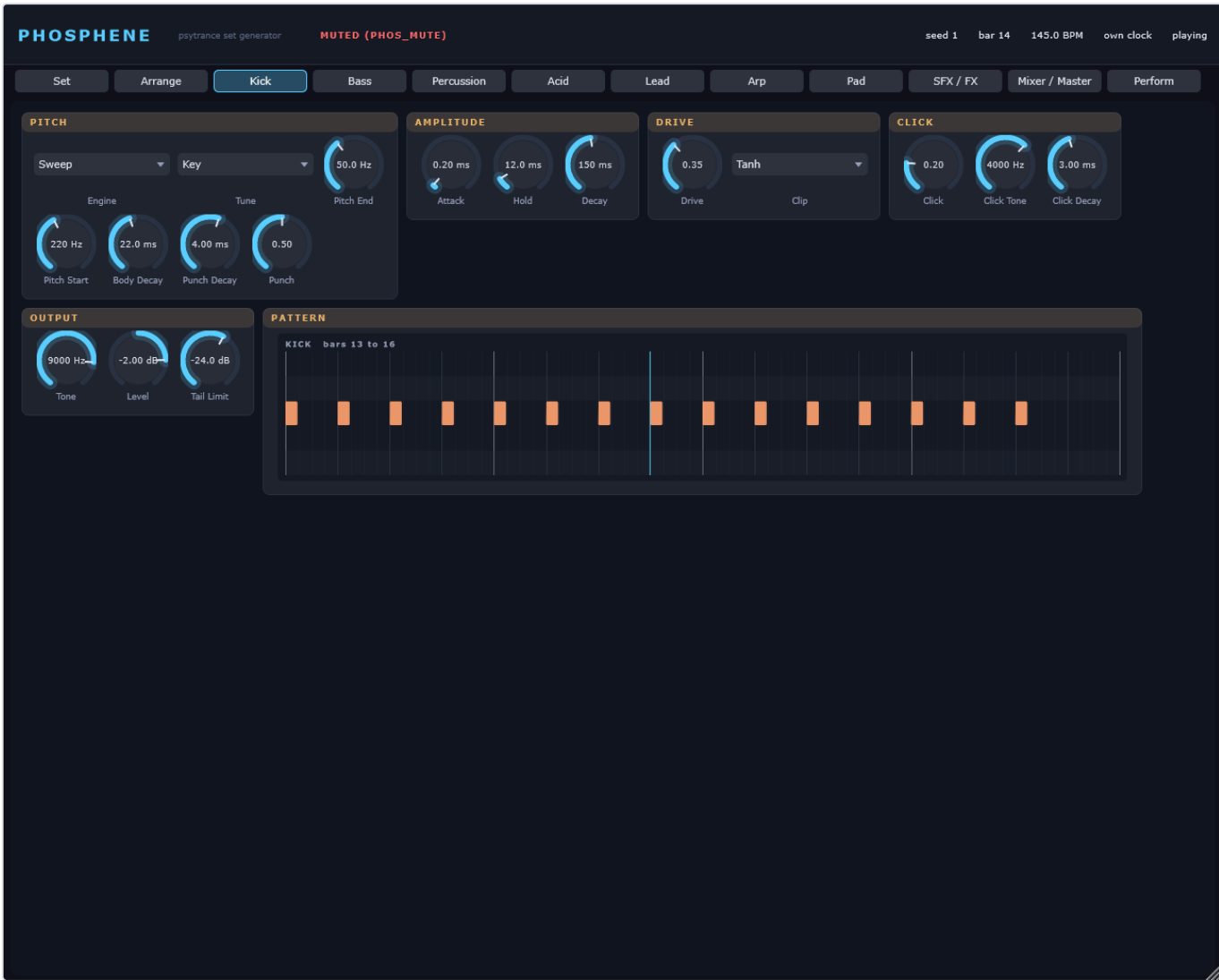
bar 14 of the set - track 1

The Arrange tab.

4. Kick

The kick is the floor of the genre and the first thing to get right. Two engines: a swept sine whose pitch falls from Pitch Start to Pitch End with its own decay, and a damped phasor resonator with the retrigger behaviour of a TR-808. Punch is a second, much faster pitch envelope on top of the first; Click adds the attack transient; Drive and Clip decide how much of the body survives.

Two of its numbers are not free. The tail is limited so that the kick has fallen far enough by the time the first bass note starts, and the phase lock (on the Bass tab) can trim the kick's decay so that the kick and the bass fundamental meet in phase. Both are applied in the engine, after every knob and every variation.



The Kick tab.

PITCH

Control	Key	Range	Default
Engine	kick.engine	Sweep Resonant	Sweep
Tune	kick.tune	Free Key	Key
Pitch End	kick.pitch_end	30 .. 120 Hz	50
Pitch Start	kick.pitch_start	60 .. 1500 Hz	220
Body Decay	kick.pitch_decay	5 .. 150 ms	22

Punch Decay	kick.punch_decay	0.5 .. 20 ms	4
Punch	kick.punch	0 .. 1	0.5

AMPLITUDE

Control	Key	Range	Default
Attack	kick.amp_attack	0 .. 10 ms	0.2
Hold	kick.amp_hold	0 .. 150 ms	12
Decay	kick.amp_decay	20 .. 1500 ms	150

DRIVE

Control	Key	Range	Default
Drive	kick.drive	0 .. 1	0.35
Clip	kick.clip	Tanh Hard	Tanh

CLICK

Control	Key	Range	Default
Click	kick.click_level	0 .. 1	0.2
Click Tone	kick.click_tone	500 .. 12000 Hz	4000
Click Decay	kick.click_decay	0.5 .. 30 ms	3

OUTPUT

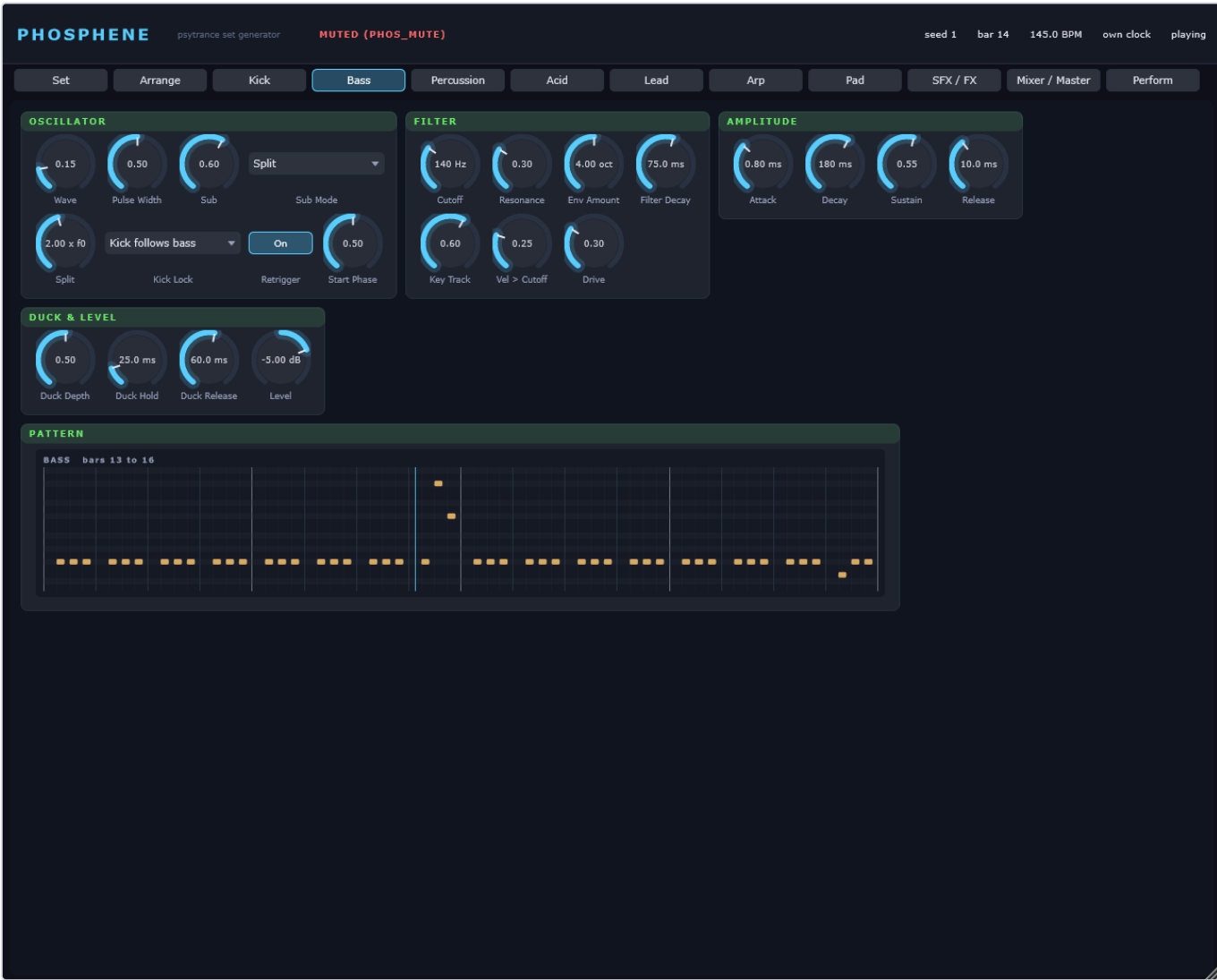
Control	Key	Range	Default
Tone	kick.tone	200 .. 20000 Hz	9000
Level	kick.level	-36 .. 6 dB	-2
Tail Limit	kick.tail_limit	-60 .. 0 dB	-24

5. Bass

The rolling bass and the kick are one instrument with two halves. The oscillator is a PolyBLEP saw and pulse with a sine sub on the fundamental -- on the fundamental, not an octave down, because the octave-down sub fought the kick -- and the filter is a non-linear zero-delay ladder.

Kick Lock decides which of the two gives way: Bass follows kick sets the bass start phase to the kick's phase at that instant, Kick follows bass trims the kick's decay by bisection until the two agree. Off leaves them where they fall, which at 145 BPM means anything between -157 and +165 degrees.

Pattern and Gate are the note side: which of the five bass patterns the track prefers, and how long a note is as a fraction of its slot. The gate is never allowed to run shorter than half a period of the fundamental.



The Bass tab.

OSCILLATOR

Control	Key	Range	Default
Wave	bass.wave	0 .. 1	0.15
Pulse Width	bass.pulse_width	0.05 .. 0.95	0.5
Sub	bass.sub	0 .. 1	0.6
Sub Mode	bass.sub_mode	Mixed Split	Split

Split	bass.split_ratio	1.2 .. 3 x f0	2
Kick Lock	bass.kick_lock	Off Bass follows kick Kick follows bass	Kick follows bass
Retrigger	bass.retrigger	off / on	on
Start Phase	bass.start_phase	0 .. 1	0.5

FILTER

Control	Key	Range	Default
Cutoff	bass.cutoff	20 .. 10000 Hz	140
Resonance	bass.resonance	0 .. 1	0.3
Env Amount	bass.env_amount	0 .. 8 oct	4
Filter Decay	bass.filter_decay	3 .. 1000 ms	75
Key Track	bass.key_track	0 .. 1	0.6
Vel > Cutoff	bass.vel_to_cutoff	0 .. 1	0.25
Drive	bass.drive	0 .. 1	0.3

AMPLITUDE

Control	Key	Range	Default
Attack	bass.amp_attack	0.1 .. 50 ms	0.8
Decay	bass.amp_decay	5 .. 2000 ms	180
Sustain	bass.amp_sustain	0 .. 1	0.55
Release	bass.amp_release	1 .. 500 ms	10

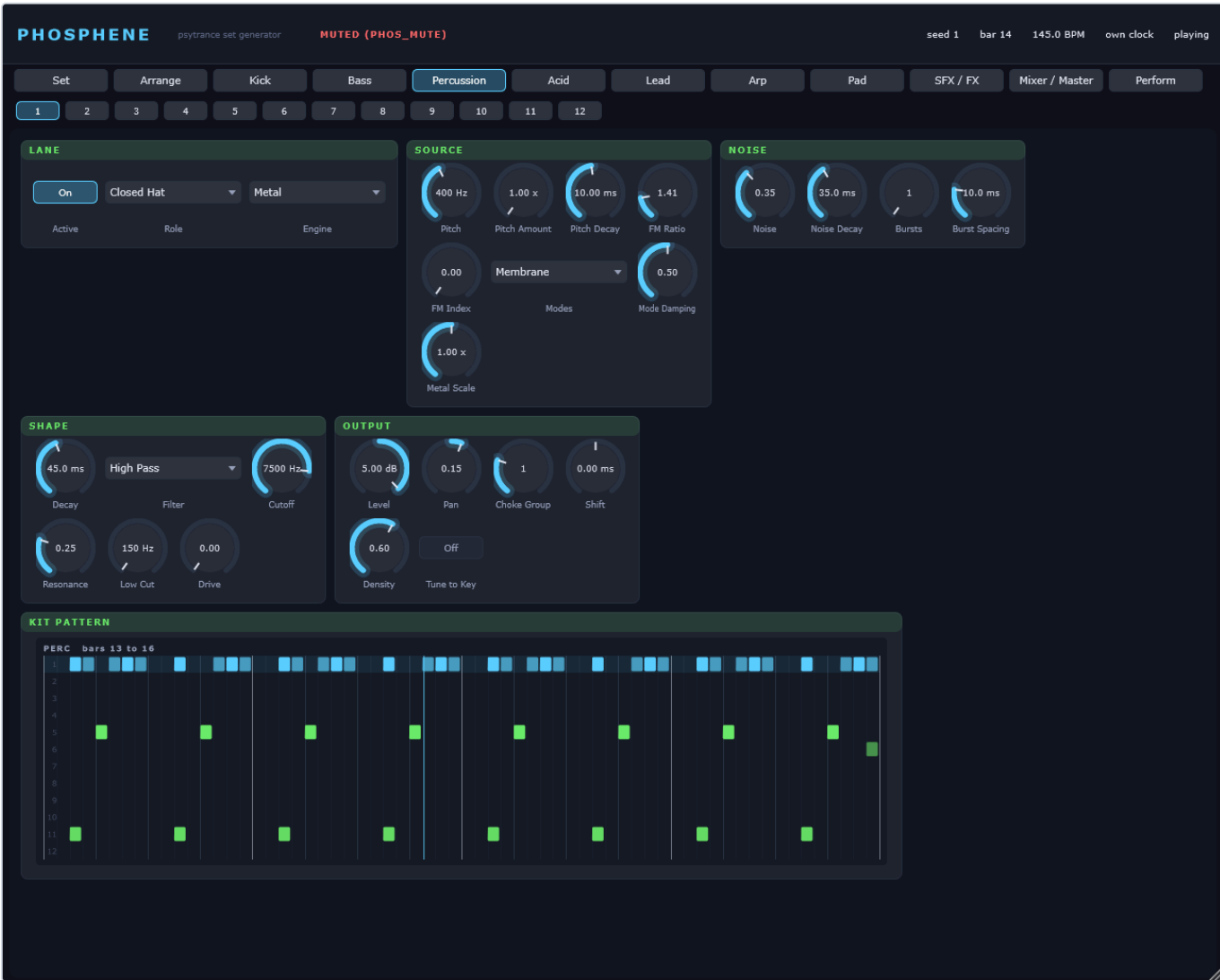
DUCK & LEVEL

Control	Key	Range	Default
Duck Depth	bass.duck_depth	0 .. 1	0.5
Duck Hold	bass.duck_hold	0 .. 200 ms	25
Duck Release	bass.duck_release	5 .. 500 ms	60
Level	bass.level	-36 .. 6 dB	-5

6. Percussion

Twelve lanes, each a complete voice: a tone with its own pitch envelope and FM, four modal sets, the 808 square stack, noise with clap bursts, a filter, a low cut, drive, choke and panning. The role of a lane decides what it plays in the groove and which General MIDI note it exports as.

The patterns are euclidean, rotated to a middle amount of syncopation rather than to none, and the decisions are made per four-bar phrase. Every lane is high-passed at 150 Hz at the least: the bottom of the mix belongs to the kick and the bass, and a tom that reaches into it takes the floor away.



The Percussion tab.

The 12 lanes of the kit share one table; lane 1 is printed here and the others are the same controls under the keys perc2.* to perc12.*.

LANE

Control	Key	Range	Default
Active	perc1.active	off / on	on
Role	perc1.role	Closed Hat Open Hat Ride Crash Clap Snare Rim Shaker Tom Conga Zap Blip	Closed Hat
Engine	perc1.engine	Noise Metal Modal Tone FM	Metal

SOURCE

Control	Key	Range	Default
Pitch	perc1.pitch	40 .. 12000 Hz	400
Pitch Amount	perc1.pitch_amount	1 .. 16 x	1
Pitch Decay	perc1.pitch_decay	0.5 .. 300 ms	10
FM Ratio	perc1.fm_ratio	0.25 .. 8	1.41
FM Index	perc1.fm_index	0 .. 8	0
Modes	perc1.mode_set	Membrane Bar Harmonic	Membrane
Mode Damping	perc1.mode_damp	0 .. 1	0.5
Metal Scale	perc1.metal_scale	0.25 .. 4 x	1

NOISE

Control	Key	Range	Default
Noise	perc1.noise	0 .. 1	0.35
Noise Decay	perc1.noise_decay	2 .. 3000 ms	35
Bursts	perc1.bursts	1 .. 6	1
Burst Spacing	perc1.burst_spacing	2 .. 40 ms	10

SHAPE

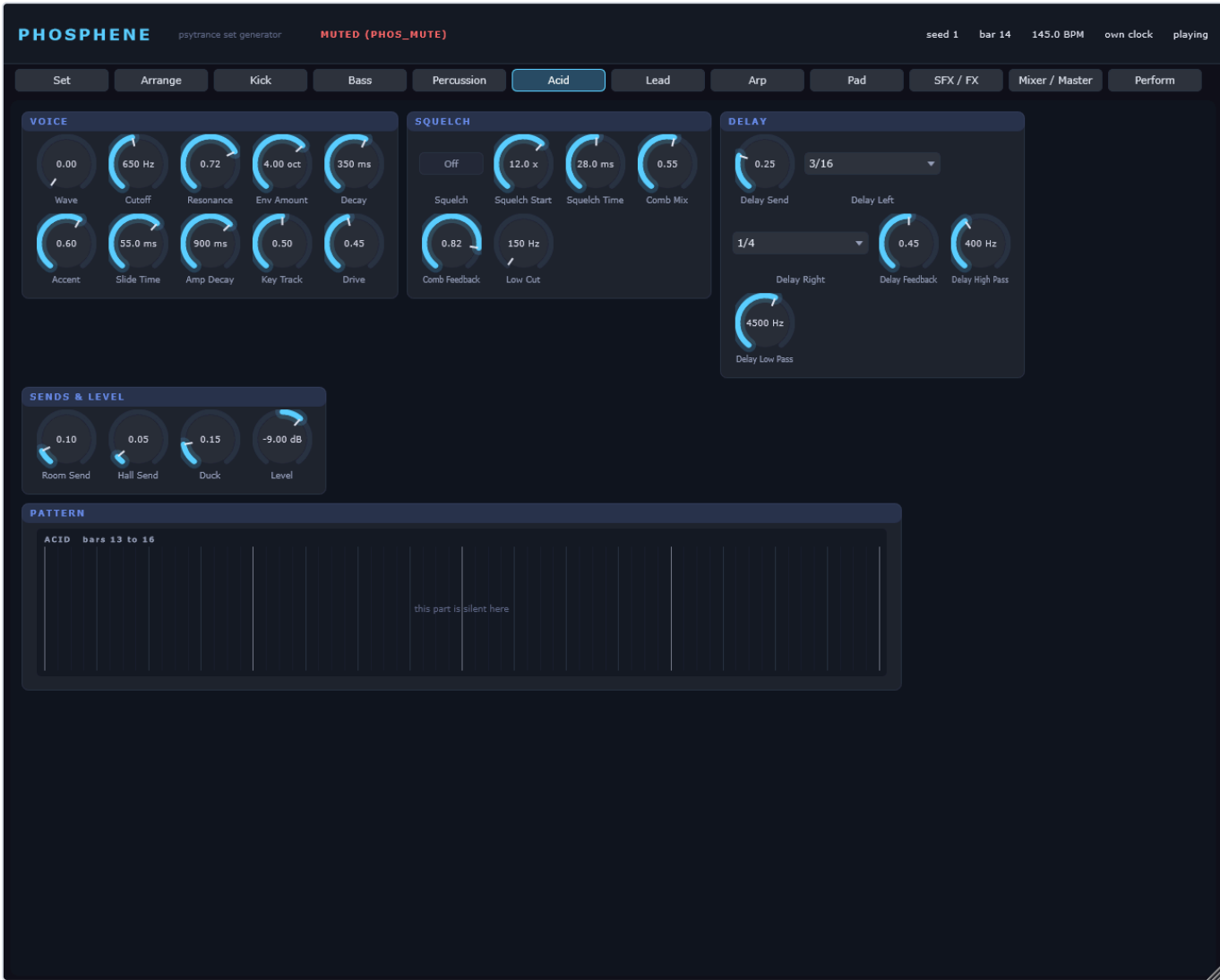
Control	Key	Range	Default
Decay	perc1.decay	2 .. 3000 ms	45
Filter	perc1.filter	Low Pass Band Pass High Pass	High Pass
Cutoff	perc1.cutoff	100 .. 18000 Hz	7500
Resonance	perc1.resonance	0 .. 1	0.25
Low Cut	perc1.low_cut	150 .. 4000 Hz	150
Drive	perc1.drive	0 .. 1	0

OUTPUT

Control	Key	Range	Default
Level	perc1.level	-36 .. 6 dB	5
Pan	perc1.pan	-1 .. 1	0.15
Choke Group	perc1.choke	0 .. 4	1
Shift	perc1.shift	-10 .. 10 ms	0
Density	perc1.density	0 .. 1	0.6
Tune to Key	perc1.tune	off / on	off

7. Acid

A diode ladder after Zavalishin, solved from the inside out, with an accent that charges a sweep capacitor the way the original did -- three accents in a row open further than one. Slide is an exponential glide into the next note, Squelch a cutoff start value plus a comb tuned to the note's period. The tempo delay is in beats, so it follows the set.



The Acid tab.

VOICE

Control	Key	Range	Default
Wave	acid.wave	0 .. 1	0
Cutoff	acid.cutoff	80 .. 8000 Hz	650
Resonance	acid.resonance	0 .. 1	0.72
Env Amount	acid.env_amount	0 .. 6 oct	4
Decay	acid.decay	30 .. 2000 ms	350
Accent	acid.accent	0 .. 1	0.6
Slide Time	acid.slide_time	5 .. 200 ms	55
Amp Decay	acid.amp_decay	50 .. 4000 ms	900
Key Track	acid.key_track	0 .. 1	0.5

Drive	acid.drive	0 .. 1	0.45
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SQUELCH

Control	Key	Range	Default
Squelch	acid.squelch	off / on	off
Squelch Start	acid.squelch_start	2 .. 32 x	12
Squelch Time	acid.squelch_time	5 .. 150 ms	28
Comb Mix	acid.comb_mix	0 .. 1	0.55
Comb Feedback	acid.comb_feedback	0 .. 0.97	0.82
Low Cut	acid.low_cut	150 .. 1000 Hz	150

DELAY

Control	Key	Range	Default
Delay Send	acid.delay_send	0 .. 1	0.25
Delay Left	acid.delay_left	1/16 1/8 3/16 1/4 3/8 1/2	3/16
Delay Right	acid.delay_right	1/16 1/8 3/16 1/4 3/8 1/2	1/4
Delay Feedback	acid.delay_feedback	0 .. 0.9	0.45
Delay High Pass	acid.delay_high_pass	150 .. 2000 Hz	400
Delay Low Pass	acid.delay_low_pass	800 .. 16000 Hz	4500

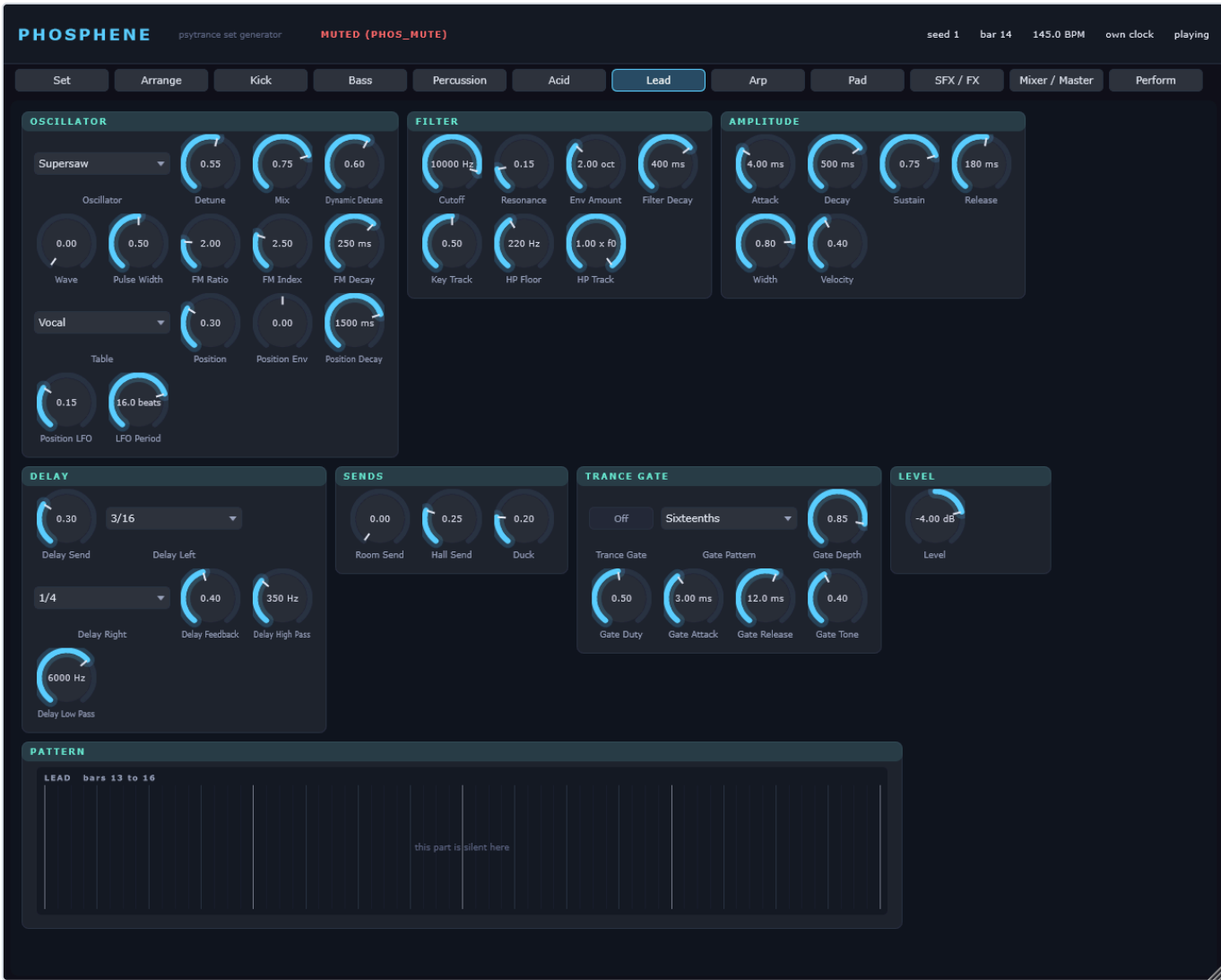
SENDS & LEVEL

Control	Key	Range	Default
Room Send	acid.room_send	0 .. 1	0.1
Hall Send	acid.hall_send	0 .. 1	0.05
Duck	acid.duck	0 .. 1	0.15
Level	acid.level	-36 .. 6 dB	-9

8. Lead

A supersaw after Szabo's measurements of the JP-8000 -- the detune curve, the mix law and the random phase per note -- with a dynamic detune that follows the length of the note. The other three oscillator types are a virtual-analogue saw and pulse, two-operator FM and a wavetable.

Everything but the kick and the bass is high-passed at its own fundamental, which is what keeps eight voices of supersaw out of the bass register.



The Lead tab.

OSCILLATOR

Control	Key	Range	Default
Oscillator	lead.osc	Supersaw VA FM Wavetable	Supersaw
Detune	lead.detune	0 .. 1	0.55
Mix	lead.mix	0 .. 1	0.75
Dynamic Detune	lead.dynamic_detune	0 .. 1	0.6
Wave	lead.wave	0 .. 1	0
Pulse Width	lead.pulse_width	0.05 .. 0.95	0.5
FM Ratio	lead.fm_ratio	0.5 .. 8	2

FM Index	lead.fm_index	0 .. 10	2.5
FM Decay	lead.fm_decay	5 .. 2000 ms	250
Table	lead.table	Classic Vocal Glass PWM Sync Formant Saw	Vocal
Position	lead.position	0 .. 1	0.3
Position Env	lead.pos_env	-1 .. 1	0
Position Decay	lead.pos_decay	10 .. 8000 ms	1500
Position LFO	lead.pos_lfo_depth	0 .. 0.5	0.15
LFO Period	lead.pos_lfo_beats	0.25 .. 64 beats	16

FILTER

Control	Key	Range	Default
Cutoff	lead.cutoff	200 .. 18000 Hz	10000
Resonance	lead.resonance	0 .. 1	0.15
Env Amount	lead.env_amount	0 .. 6 oct	2
Filter Decay	lead.filter_decay	5 .. 3000 ms	400
Key Track	lead.key_track	0 .. 1	0.5
HP Floor	lead.hp_floor	150 .. 400 Hz	220
HP Track	lead.hp_track	0 .. 1 x f0	1

AMPLITUDE

Control	Key	Range	Default
Attack	lead.amp_attack	0.3 .. 2000 ms	4
Decay	lead.amp_decay	5 .. 4000 ms	500
Sustain	lead.amp_sustain	0 .. 1	0.75
Release	lead.amp_release	5 .. 4000 ms	180
Width	lead.width	0 .. 1	0.8
Velocity	lead.vel_sens	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	lead.delay_send	0 .. 1	0.3
Delay Left	lead.delay_left	1/16 1/8 3/16 1/4 3/8 1/2	3/16
Delay Right	lead.delay_right	1/16 1/8 3/16 1/4 3/8 1/2	1/4
Delay Feedback	lead.delay_feedback	0 .. 0.9	0.4
Delay High Pass	lead.delay_high_pass	150 .. 2000 Hz	350
Delay Low Pass	lead.delay_low_pass	800 .. 16000 Hz	6000

SENDS

Control	Key	Range	Default
Room Send	lead.room_send	0 .. 1	0
Hall Send	lead.hall_send	0 .. 1	0.25

Duck	lead.duck	0 .. 1	0.2
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TRANCE GATE

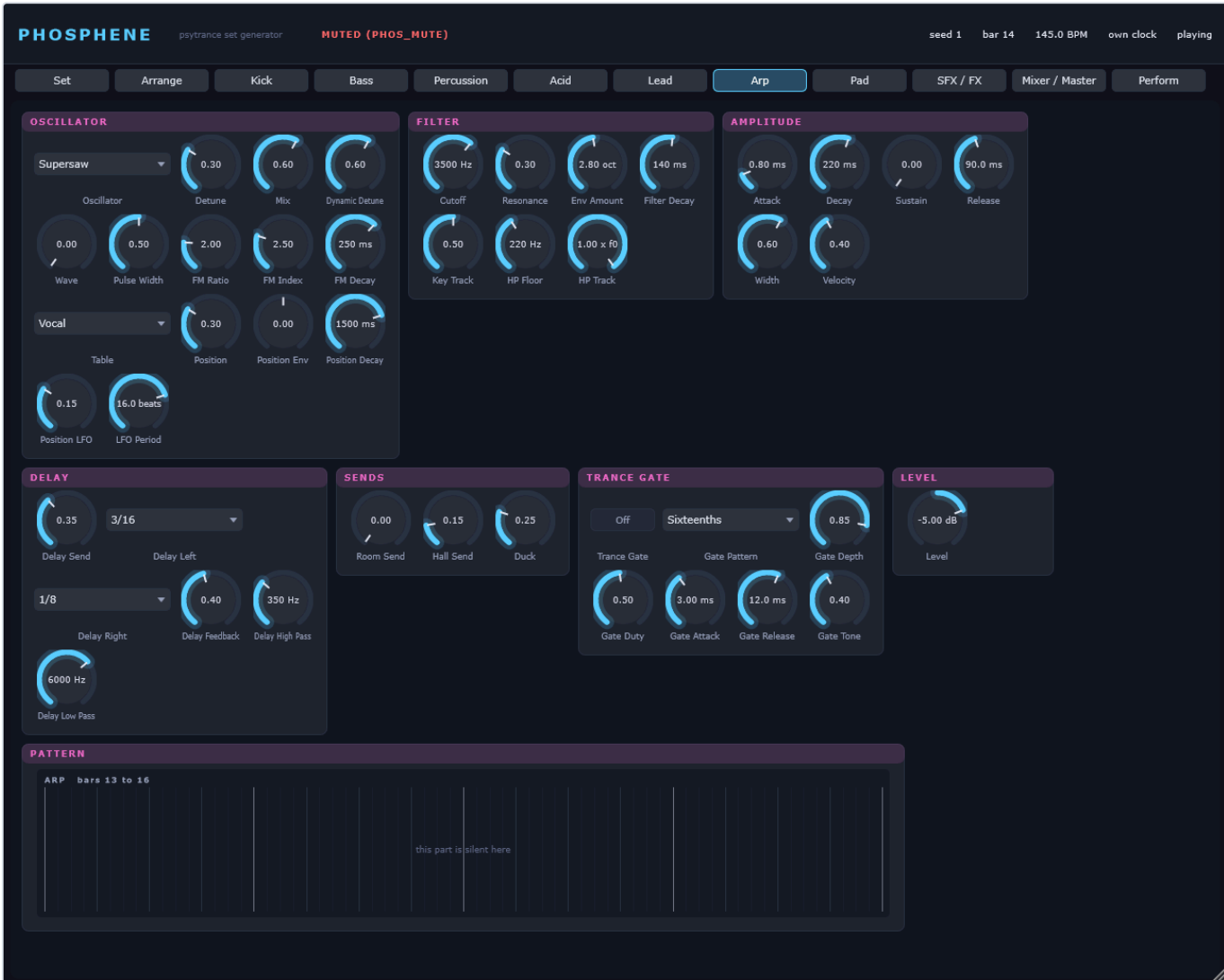
Control	Key	Range	Default
Trance Gate	lead.gate	off / on	off
Gate Pattern	lead.gate_pattern	Sixteenths Eighths Rolling Gallop 3-3-2 Triplets	Sixteenths
Gate Depth	lead.gate_depth	0 .. 1	0.85
Gate Duty	lead.gate_duty	0.05 .. 1	0.5
Gate Attack	lead.gate_attack	0.5 .. 60 ms	3
Gate Release	lead.gate_release	0.5 .. 120 ms	12
Gate Tone	lead.gate_tone	0 .. 1	0.4

LEVEL

Control	Key	Range	Default
Level	lead.level	-36 .. 6 dB	-4

9. Arp

The same engine as the lead, with its own knobs. The arpeggio runs on chord tones, and when its pitch range would overlap the lead's by more than two semitones it is octaved away: two voices playing in the same two octaves mask each other, and the composer checks that before it writes the notes.



The Arp tab.

OSCILLATOR

Control	Key	Range	Default
Oscillator	arp.osc	Supersaw VA FM Wavetable	Supersaw
Detune	arp.detune	0 .. 1	0.3
Mix	arp.mix	0 .. 1	0.6
Dynamic Detune	arp.dynamic_detune	0 .. 1	0.6
Wave	arp.wave	0 .. 1	0
Pulse Width	arp.pulse_width	0.05 .. 0.95	0.5
FM Ratio	arp.fm_ratio	0.5 .. 8	2
FM Index	arp.fm_index	0 .. 10	2.5
FM Decay	arp.fm_decay	5 .. 2000 ms	250

Table	arp.table	Classic Vocal Glass PWM Sync Formant Saw	Vocal
Position	arp.position	0 .. 1	0.3
Position Env	arp.pos_env	-1 .. 1	0
Position Decay	arp.pos_decay	10 .. 8000 ms	1500
Position LFO	arp.pos_lfo_depth	0 .. 0.5	0.15
LFO Period	arp.pos_lfo_beats	0.25 .. 64 beats	16

FILTER

Control	Key	Range	Default
Cutoff	arp.cutoff	200 .. 18000 Hz	3500
Resonance	arp.resonance	0 .. 1	0.3
Env Amount	arp.env_amount	0 .. 6 oct	2.8
Filter Decay	arp.filter_decay	5 .. 3000 ms	140
Key Track	arp.key_track	0 .. 1	0.5
HP Floor	arp.hp_floor	150 .. 400 Hz	220
HP Track	arp.hp_track	0 .. 1 x f0	1

AMPLITUDE

Control	Key	Range	Default
Attack	arp.amp_attack	0.3 .. 2000 ms	0.8
Decay	arp.amp_decay	5 .. 4000 ms	220
Sustain	arp.amp_sustain	0 .. 1	0
Release	arp.amp_release	5 .. 4000 ms	90
Width	arp.width	0 .. 1	0.6
Velocity	arp.vel_sens	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	arp.delay_send	0 .. 1	0.35
Delay Left	arp.delay_left	1/16 1/8 3/16 1/4 3/8 1/2	3/16
Delay Right	arp.delay_right	1/16 1/8 3/16 1/4 3/8 1/2	1/8
Delay Feedback	arp.delay_feedback	0 .. 0.9	0.4
Delay High Pass	arp.delay_high_pass	150 .. 2000 Hz	350
Delay Low Pass	arp.delay_low_pass	800 .. 16000 Hz	6000

SENDS

Control	Key	Range	Default
Room Send	arp.room_send	0 .. 1	0
Hall Send	arp.hall_send	0 .. 1	0.15
Duck	arp.duck	0 .. 1	0.25

TRANCE GATE

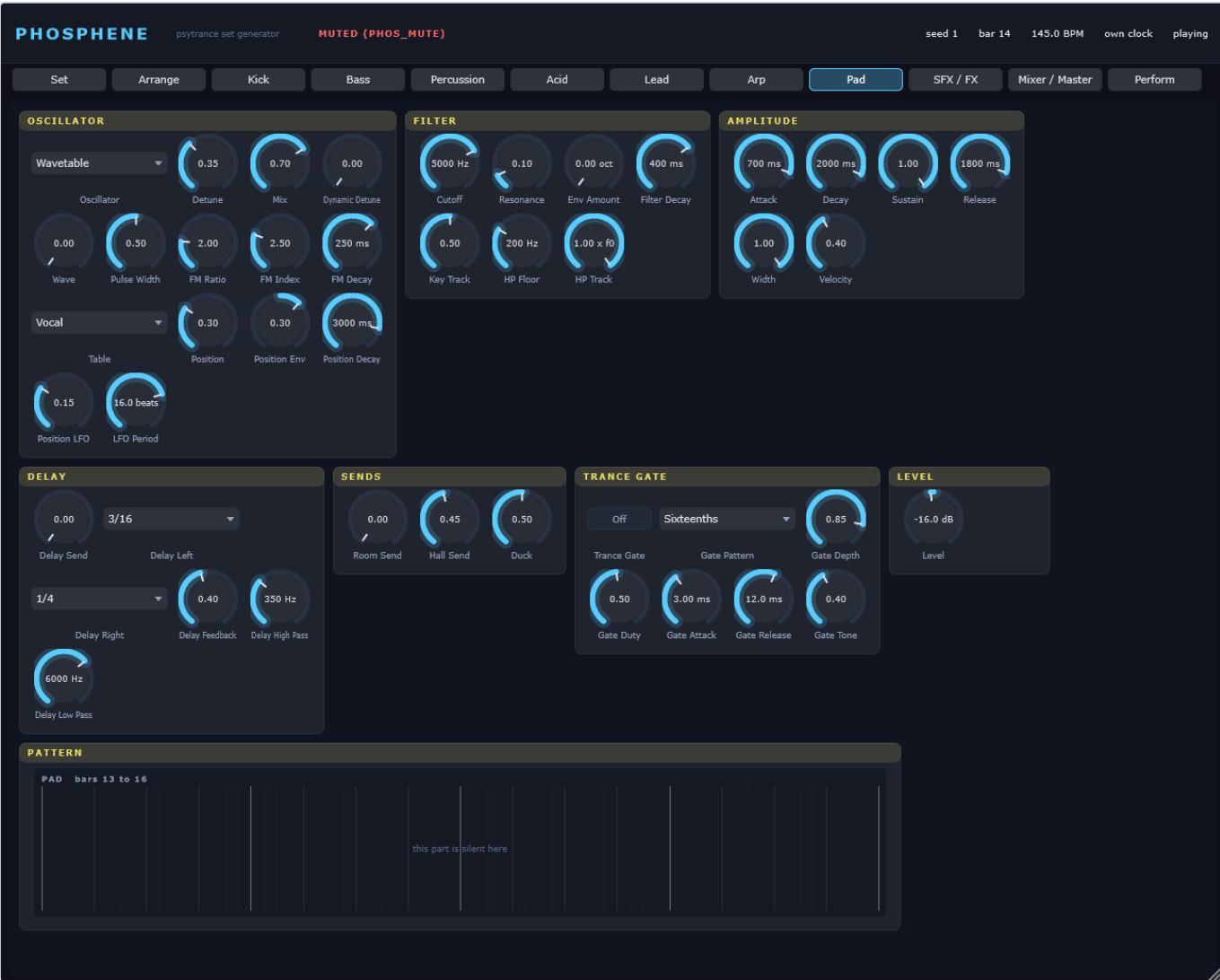
Control	Key	Range	Default
Trance Gate	arp.gate	off / on	off
Gate Pattern	arp.gate_pattern	Sixteenths Eighths Rolling Gallop 3-3-2 Triplets	Sixteenths
Gate Depth	arp.gate_depth	0 .. 1	0.85
Gate Duty	arp.gate_duty	0.05 .. 1	0.5
Gate Attack	arp.gate_attack	0.5 .. 60 ms	3
Gate Release	arp.gate_release	0.5 .. 120 ms	12
Gate Tone	arp.gate_tone	0 .. 1	0.4

LEVEL

Control	Key	Range	Default
Level	arp.level	-36 .. 6 dB	-5

10. Pad

Four-voice voicings with the smallest voice movement between chords, through the wavetable engine. The pads carry the blocks that have no lead and no acid, and the trance gate gives them their movement.



The Pad tab.

OSCILLATOR

Control	Key	Range	Default
Oscillator	pad.osc	Supersaw VA FM Wavetable	Wavetable
Detune	pad.detune	0 .. 1	0.35
Mix	pad.mix	0 .. 1	0.7
Dynamic Detune	pad.dynamic_detune	0 .. 1	0
Wave	pad.wave	0 .. 1	0
Pulse Width	pad.pulse_width	0.05 .. 0.95	0.5
FM Ratio	pad.fm_ratio	0.5 .. 8	2
FM Index	pad.fm_index	0 .. 10	2.5
FM Decay	pad.fm_decay	5 .. 2000 ms	250
Table	pad.table	Classic Vocal Glass PWM Sync Formant Saw	Vocal

Position	pad.position	0 .. 1	0.3
Position Env	pad.pos_env	-1 .. 1	0.3
Position Decay	pad.pos_decay	10 .. 8000 ms	3000
Position LFO	pad.pos_lfo_depth	0 .. 0.5	0.15
LFO Period	pad.pos_lfo_beats	0.25 .. 64 beats	16

FILTER

Control	Key	Range	Default
Cutoff	pad.cutoff	200 .. 18000 Hz	5000
Resonance	pad.resonance	0 .. 1	0.1
Env Amount	pad.env_amount	0 .. 6 oct	0
Filter Decay	pad.filter_decay	5 .. 3000 ms	400
Key Track	pad.key_track	0 .. 1	0.5
HP Floor	pad.hp_floor	150 .. 400 Hz	200
HP Track	pad.hp_track	0 .. 1 x f0	1

AMPLITUDE

Control	Key	Range	Default
Attack	pad.amp_attack	0.3 .. 2000 ms	700
Decay	pad.amp_decay	5 .. 4000 ms	2000
Sustain	pad.amp_sustain	0 .. 1	1
Release	pad.amp_release	5 .. 4000 ms	1800
Width	pad.width	0 .. 1	1
Velocity	pad.vel_sens	0 .. 1	0.4

DELAY

Control	Key	Range	Default
Delay Send	pad.delay_send	0 .. 1	0
Delay Left	pad.delay_left	1/16 1/8 3/16 1/4 3/8 1/2	3/16
Delay Right	pad.delay_right	1/16 1/8 3/16 1/4 3/8 1/2	1/4
Delay Feedback	pad.delay_feedback	0 .. 0.9	0.4
Delay High Pass	pad.delay_high_pass	150 .. 2000 Hz	350
Delay Low Pass	pad.delay_low_pass	800 .. 16000 Hz	6000

SENDS

Control	Key	Range	Default
Room Send	pad.room_send	0 .. 1	0
Hall Send	pad.hall_send	0 .. 1	0.45
Duck	pad.duck	0 .. 1	0.5

TRANCE GATE

Control	Key	Range	Default
Trance Gate	pad.gate	off / on	off
Gate Pattern	pad.gate_pattern	Sixteenths Eighths Rolling Gallop 3-3-2 Triplets	Sixteenths
Gate Depth	pad.gate_depth	0 .. 1	0.85
Gate Duty	pad.gate_duty	0.05 .. 1	0.5
Gate Attack	pad.gate_attack	0.5 .. 60 ms	3
Gate Release	pad.gate_release	0.5 .. 120 ms	12
Gate Tone	pad.gate_tone	0 .. 1	0.4

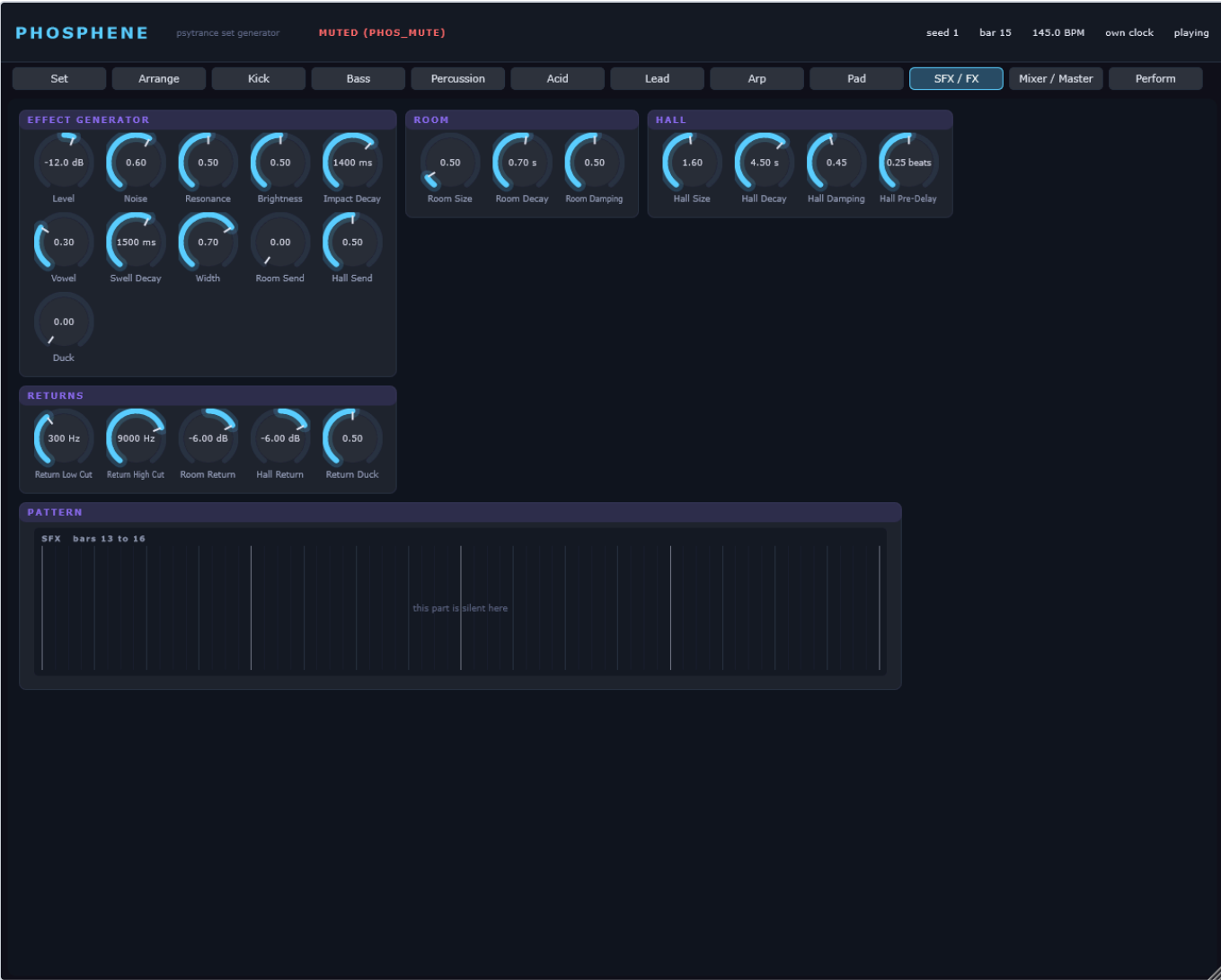
LEVEL

Control	Key	Range	Default
Level	pad.level	-36 .. 6 dB	-16

11. SFX / FX

The effect generator makes risers, downlifters, impacts, sweeps, formant shots, reverse swells and zaps; the composer places them at the form's boundaries -- a riser over the last eight bars of a buildup, ending exactly on the drop, the formant shot on the last beat of the pre-drop break, a descending sweep into the drop.

The two send effects are a short room and a long hall, both feedback delay networks. Their returns are ducked with the kick like everything else, and high-passed: a reverb tail in the bass register is mud.



The SFX / FX tab.

EFFECT GENERATOR

Control	Key	Range	Default
Level	<code>sfx.level</code>	-36 .. 6 dB	-12
Noise	<code>sfx.noise</code>	0 .. 1	0.6
Resonance	<code>sfx.resonance</code>	0 .. 1	0.5
Brightness	<code>sfx.brightness</code>	0 .. 1	0.5
Impact Decay	<code>sfx.impact_decay</code>	200 .. 4000 ms	1400
Vowel	<code>sfx.vowel</code>	0 .. 1	0.3
Swell Decay	<code>sfx.swell_decay</code>	200 .. 6000 ms	1500

Width	sfx.width	0 .. 1	0.7
Room Send	sfx.room_send	0 .. 1	0
Hall Send	sfx.hall_send	0 .. 1	0.5
Duck	sfx.duck	0 .. 1	0

ROOM

Control	Key	Range	Default
Room Size	fx.room_size	0.3 .. 3	0.5
Room Decay	fx.room_decay	0.1 .. 4 s	0.7
Room Damping	fx.room_damping	0 .. 1	0.5

HALL

Control	Key	Range	Default
Hall Size	fx.hall_size	0.3 .. 3	1.6
Hall Decay	fx.hall_decay	0.3 .. 20 s	4.5
Hall Damping	fx.hall_damping	0 .. 1	0.45
Hall Pre-Delay	fx.hall_pre_delay	0 .. 0.5 beats	0.25

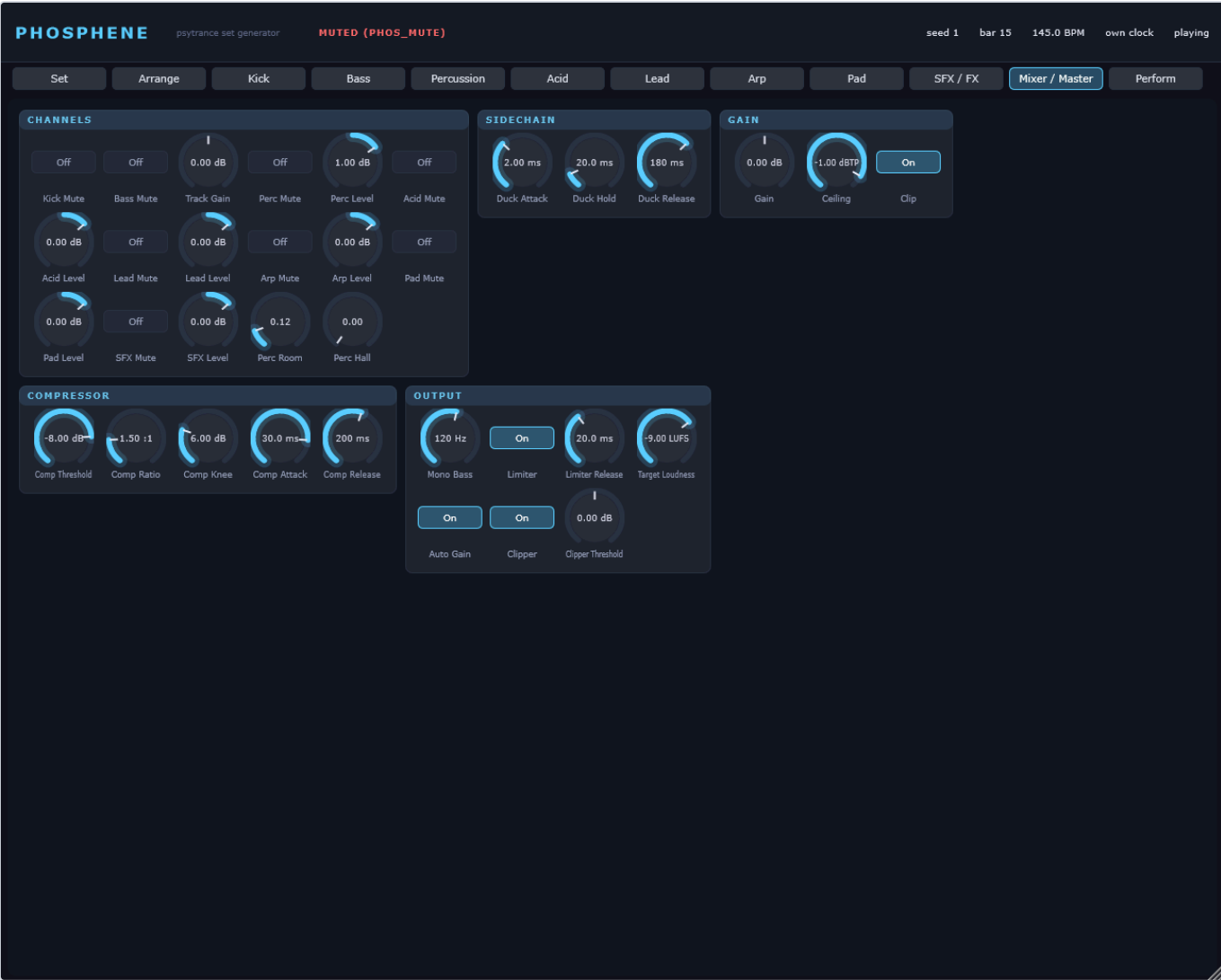
RETURNS

Control	Key	Range	Default
Return Low Cut	fx.low_cut	150 .. 1000 Hz	300
Return High Cut	fx.high_cut	1000 .. 20000 Hz	9000
Room Return	fx.room_return	-36 .. 6 dB	-6
Hall Return	fx.hall_return	-36 .. 6 dB	-6
Return Duck	fx.return_duck	0 .. 1	0.5

12. Mixer / Master

Level, mute and sends per part, the sidechain matrix, and the master chain: bus compressor, mono bass, a two-times oversampled knee clipper, a lookahead true-peak limiter and a safety clip, metered to ITU-R BS.1770.

Auto Gain renders four four-bar windows of the track through the whole master chain, measures them and offsets the master gain so that the output meets the loudness target. It runs when a track is planned, so it uses the knobs as they stand at that moment.



The Mixer / Master tab.

CHANNELS

Control	Key	Range	Default
Kick Mute	<code>mix.kick_mute</code>	off / on	off
Bass Mute	<code>mix.bass_mute</code>	off / on	off
Track Gain	<code>mix.track_gain</code>	-12 .. 12 dB	0
Perc Mute	<code>mix.perc_mute</code>	off / on	off
Perc Level	<code>mix.perc_level</code>	-24 .. 12 dB	1
Acid Mute	<code>mix.acid_mute</code>	off / on	off
Acid Level	<code>mix.acid_level</code>	-24 .. 12 dB	0

Lead Mute	mix.lead_mute	off / on	off
Lead Level	mix.lead_level	-24 .. 12 dB	0
Arp Mute	mix.arp_mute	off / on	off
Arp Level	mix.arp_level	-24 .. 12 dB	0
Pad Mute	mix.pad_mute	off / on	off
Pad Level	mix.pad_level	-24 .. 12 dB	0
SFX Mute	mix.sfx_mute	off / on	off
SFX Level	mix.sfx_level	-24 .. 12 dB	0
Perc Room	mix.perc_room	0 .. 1	0.12
Perc Hall	mix.perc_hall	0 .. 1	0

SIDECHAIN

Control	Key	Range	Default
Duck Attack	mix.duck_attack	0.5 .. 30 ms	2
Duck Hold	mix.duck_hold	0 .. 200 ms	20
Duck Release	mix.duck_release	10 .. 800 ms	180

GAIN

Control	Key	Range	Default
Gain	master.gain	-24 .. 24 dB	0
Ceiling	master.ceiling	-12 .. 0 dBTP	-1
Clip	master.clip	off / on	on

COMPRESSOR

Control	Key	Range	Default
Comp Threshold	master.comp_threshold	-40 .. 0 dB	-8
Comp Ratio	master.comp_ratio	1 .. 10 :1	1.5
Comp Knee	master.comp_knee	0 .. 24 dB	6
Comp Attack	master.comp_attack	0.1 .. 100 ms	30
Comp Release	master.comp_release	10 .. 2000 ms	200

OUTPUT

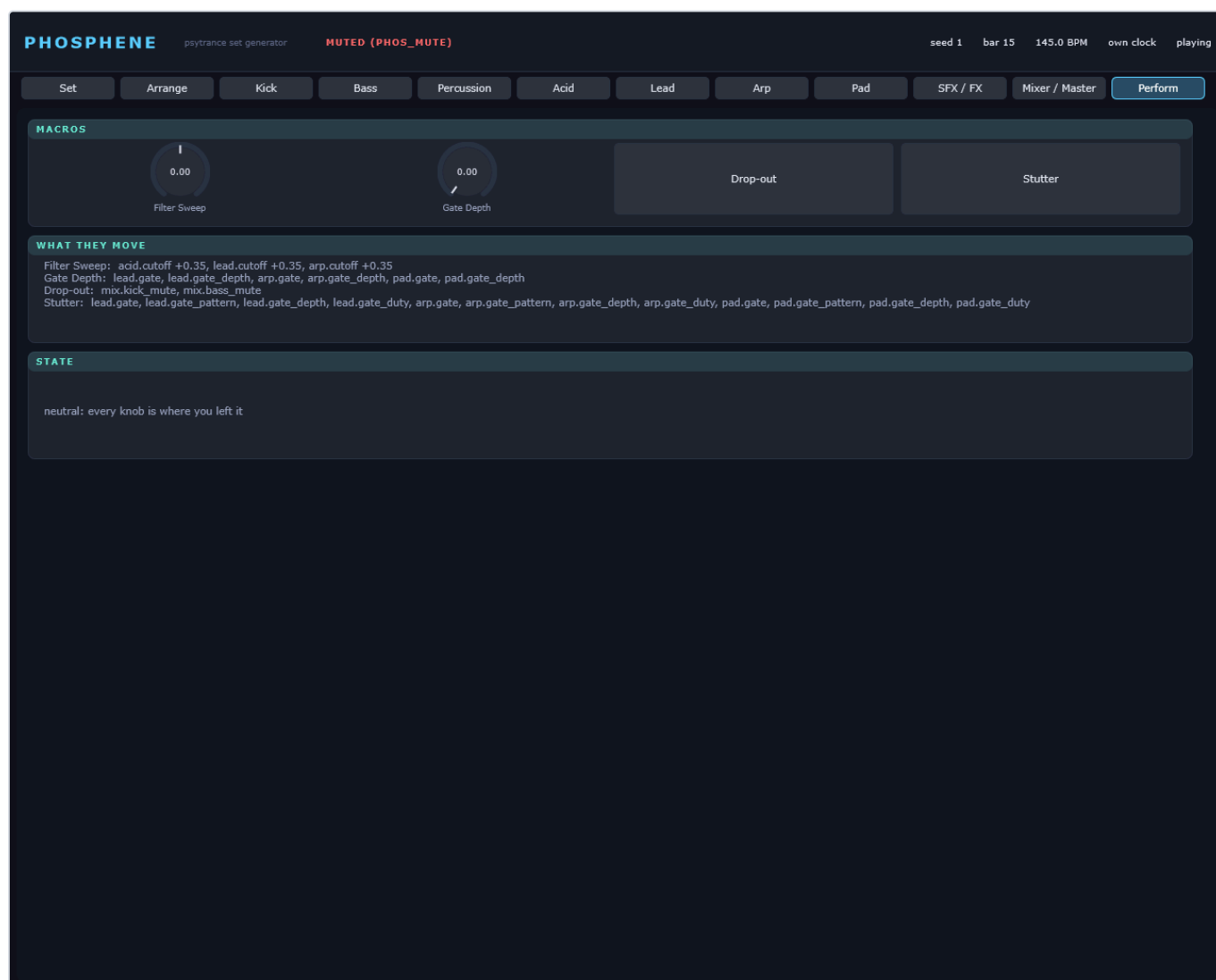
Control	Key	Range	Default
Mono Bass	master.mono_bass	40 .. 300 Hz	120
Limiter	master.limiter	off / on	on
Limiter Release	master.limiter_release	2 .. 1000 ms	20
Target Loudness	master.target_lufs	-20 .. -4 LUFS	-9
Auto Gain	master.auto_gain	off / on	on
Clipper	master.clipper	off / on	on
Clipper Threshold	master.clipper_threshold	-6 .. 6 dB	0

13. Perform

Four macros, meant for a hand rather than a mouse.

They are immediate parameter moves rather than score events, and that is measured rather than convenient: the engine's control ring is first in, first out and the composer keeps it filled eight bars ahead, so an event pushed into it now would not be heard for about thirteen seconds. A macro therefore writes the knob itself, as an offset in the knob's normalised domain, and puts it back exactly where it was when you let go. The composer's own ramps keep riding on top.

Drop-out is a press, not a hold: it takes the kick and the bass out and lets them back in on the next bar line. Stutter is held. Filter Sweep and Gate Depth stay where you leave them, and a double click puts them back to neutral.



The Perform tab.

Filter Sweep

Opens or closes the acid, lead and arp filters together, by up to a third of their range in each direction. Centre is neutral.

Moves: acid.cutoff +0.35, lead.cutoff +0.35, arp.cutoff +0.35

Gate Depth

Switches the trance gate on for lead, arp and pad and sets its depth; at 1.0 the gate closes completely between its steps.

Moves: `lead.gate = 1, lead.gate_depth = 1, arp.gate = 1, arp.gate_depth = 1, pad.gate = 1, pad.gate_depth = 1`

Drop-out (press)

Takes kick and bass out until the next bar line and lets them back in on the downbeat. One press, one bar.

Moves: `mix.kick_mute = 1, mix.bass_mute = 1`

Stutter (press)

Held: lead, arp and pad run through the gate on sixteenths at full depth and the shortest duty. Phosphene has no buffer repeat, so this is the gate's stutter, not a tape one.

Moves: `lead.gate = 1, lead.gate_pattern = 0, lead.gate_depth = 1, lead.gate_duty = 0, arp.gate = 1, arp.gate_pattern = 0, arp.gate_depth = 1, arp.gate_duty = 0, pad.gate = 1, pad.gate_pattern = 0, pad.gate_depth = 1, pad.gate_duty = 0`